

Harsh Shah

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EDUCATION

Northeastern University, Boston, MA Jan. 2024
Master of Science in Computer Science, **GPA: 3.8/4.0** Graduation: Dec 2025
Relevant Courses: Programming Design Paradigm, Foundations of Artificial Intelligence, Algorithms, Machine Learning

Savitribai Phule Pune University, Pune, India June 2023
Bachelor of Engineering in Computer Engineering, **GPA: 9.1/10.0**
Relevant Courses: Data Structures and Algorithms, Cloud Computing, Object Oriented Design, Big Data Processing and Analytics, Database Management Systems, Distributed Systems

SKILLS

Languages: Java, C++, Python, C#
Web Technologies: HTML, CSS, React, NodeJS, Blazor
Databases: MySQL, Microsoft SQL Server, MongoDB, DynamoDB, Snowflake
Cloud & Ops: Grafana, Prometheus, CI/CD, Docker, Jenkins, Microsoft Azure, AWS S3 Buckets
Frameworks: Git, GitHub, .NET, Kubernetes, Apache Spark, TensorFlow, Keras, PyTorch, Scikit-Learn, NLTK

WORK EXPERIENCE

Northeastern University, Boston, Massachusetts May 2024 – Present
• **Teaching Assistant** for CS 5800 (Algorithms) - Fall 2025 and CS 3000 (Algorithms and Data) - Summer-1 & Fall 2024.

Atlas SP Partners, New York City, New York Jan. 2025 – Aug. 2025
Software Engineer Intern
• Collaborated in a **cross-functional team** to develop and implement full stack applications, automation workflows and centralized database storage for the **Residential Mortgage** division of Securitized Products.
• Automated the daily margin call workflow using **Python saving 3-5 hours** of manual effort daily.
• Developed **full stack solution** for daily margin call workflow using **.NET** as **backend** and **React** as **frontend** with number of **microservices** hosted on **Azure**.
• Built a **config-driven, fully automated data ingestion pipeline** in **Python**, integrating data from **3+ businesses units** into **Snowflake**, enabling centralized, reliable storage and **reducing Excel based systems by 70%**.
• Configured and managed **automated workflow jobs** in **Control-M Helix** to automate and orchestrate daily processes.
• Developed a **Python** based fully automated solution for Resi Due Diligence, paired with an **Excel interface via PyXII**, reducing manual effort by **15-20 hours** per deal.
• Developed an **email monitoring Python package** using **Microsoft Graph API** to track mailboxes and send bulk emails.

Knorr-Bremse Technology Center India Pvt Ltd, Pune, India Apr. 2022 – Apr. 2023
Software Engineer Intern
• Improved the efficiency and accuracy of the data management process by **70%** by automating the process of MS SQL database management by developing a **Python** script.
• Designed and maintained **Microsoft SQL Server database schema** to support various Software Engineering Key Performance Indicator (KPI) metrics following the **Agile methodology** for software development.
• Configured and developed **Grafana** dashboards for data visualization of various KPI metrics by developing **SQL** queries.
• Continuous Integration and Deployment (**CI/CD**) using **Jenkins** pipelines, ensuring consistent and reliable deployments.

PROJECTS

Text Translation, Transformer and LSTM Architecture ([link](#)) Jan. 2024 – Apr. 2024
• Collaborated on a team project to develop an efficient, high-quality text translation system using **natural language processing architectures** to promote global communication and cultural exchange.
• Developed **transformer** architecture from scratch to translate from **French to English**, achieving an accuracy of **91%**.

Image Manipulation and Forgery Detection, Image Segmentation Model and Deep Learning ([link](#)) July 2022 – May 2023
• Led a machine learning project on image manipulation detection model using the **ResNet** and **Unet** architecture achieving **93%** accuracy on the **CASIA dataset**, while improving efficiency through a survey of existing methods.

Driver Drowsiness Detection, OpenCV and Convolutional Neural Network ([link](#)) July 2021 – Sept. 2021
• Developed a real-time **Python** application using **OpenCV** and **CNNs** to detect driver drowsiness with **90%** accuracy, integrating an alarm system for timely intervention and accident prevention.

PUBLICATIONS

- Exploring the Potential of Generative Adversarial Networks: A Comparative Study of GAN ([link](#))
- Autonomous UAV Swarms: Powering Up with AI ([link](#))